SQL Code

```sql
USE employee;

SELECT DISTINCT salary
FROM employees;
SELECT age    AS Employee_Age,
       salary AS Employee_Salary
FROM employees;
SELECT *
FROM employees
WHERE salary > 50000
  AND hire_date < '2016-01-01';
  SELECT *
FROM employees
WHERE designation IS NULL OR designation = '';
SET SQL_SAFE_UPDATES = 0;

UPDATE employees
SET designation = 'Data Scientist'
WHERE designation IS NULL OR designation = '';
SELECT employee_id, employee_name, designation
FROM employees
WHERE designation = 'Data Scientist';
SELECT *
FROM employees
ORDER BY department_id ASC, salary DESC;
SELECT *
FROM employees
WHERE hire_date >= '2018-01-01'
  AND hire_date <  '2019-01-01'
ORDER BY hire_date ASC
LIMIT 5;

SELECT SUM(e.salary) AS total_finance_salary
FROM employees e
JOIN departments d ON e.department_id = d.department_id
WHERE d.department_name = 'Finance';


SELECT MIN(age) AS min_age
FROM employees;

SELECT l.location,
       MAX(e.salary) AS max_salary
FROM employees e
JOIN location l ON e.location_id = l.location_id
GROUP BY l.location;

SELECT designation,
       AVG(salary) AS avg_salary
FROM employees
```

```sql
WHERE designation LIKE '%Analyst%'
GROUP BY designation;

SELECT d.department_name,
       COUNT(e.employee_id) AS employee_count
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name
HAVING COUNT(e.employee_id) < 3;

SELECT l.location,
       AVG(e.age) AS avg_age
FROM employees e
JOIN location l ON e.location_id = l.location_id
WHERE e.gender = 'F'
GROUP BY l.location
HAVING AVG(e.age) < 30;

SELECT e.employee_name,
       e.designation,
       d.department_name
FROM employees e
INNER JOIN departments d ON e.department_id = d.department_id;

SELECT d.department_name,
       COUNT(e.employee_id) AS total_employees
FROM departments d
LEFT JOIN employees e ON d.department_id = e.department_id
GROUP BY d.department_id, d.department_name;

SELECT l.location,
       e.employee_name
FROM employees e
RIGHT JOIN location l ON e.location_id = l.location_id;


```